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A Stochastic Approach for Generating 3D Digital Sandstone Models with Consistent Pore Networks and Grayscale Variability

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Abstract

Digital rocks are valuable tools for characterizing porous media and play a key role in advancing our understanding of subsurface processes. This work introduces a stochastic modeling framework for generating synthetic 3D pore networks and digital sandstone samples that reproduce key statistical and geometric patterns observed in micro-CT data, including pore connectivity and grayscale heterogeneity. The underlying stochastic model enables systematic computational experiments to investigate how network connectivity, porosity, and material heterogeneity influence pore-scale behavior and transport properties. The method employs thresholded Gaussian random fields (GRFs) to model pore networks, while the grayscale variation representing material heterogeneity is modeled using random fields statistically parameterized from imaging data and the corresponding grayscale distribution. A double-thresholding approach is proposed that distinguishes pores and clay-like transitional material, enhancing microstructural expressiveness. GRF parameters, such as correlation length, variance, and regularity of the synthetic realizations, are tuned to match sandstone features. An additional contribution is the incorporation of a morphological filtering step that further refines the structures and controls fine-scale connectivity. The fidelity of the synthetic models is validated using Minkowski functionals and distribution-based metrics, benchmarking against a high-resolution 1000³-voxel Berea sandstone sample from micro-CT. The GRF sampling is computationally efficient, making it feasible for generating synthetic datasets that support studies on larger representative volumes. The generated synthetic sandstones show encouraging agreement with structural, geometric, and material characteristics observed in the real sample. This systematic and extensible framework, with interpretable parameters, enables generation of a wide range of pore structures and heterogeneities, supporting sensitivity analysis, uncertainty quantification, and pore-scale flow simulations. It opens new avenues for extending digital rock modeling to non-homogeneous formations, including adaptations that account for heavy mineral inclusions and other lithologies.

Introduction

In digital rock analysis, X-ray micro-computed tomography (micro-CT) provides a means to evaluate rock samples at the pore scale by generating detailed 3D digital model representations. These digital twins,

in turn, help in quantifying macrostructural properties that influence flow, such as effective porosity and absolute and relative permeability, based on the underlying microstructure (Andrä et al., 2013). However, such analyses depend on a limited number of core samples and are further constrained by natural variability and imaging resolution. Synthetic digital rocks that can be stochastically generated to match targeted statistical distributions of porosity, connectivity, pore sizes, tortuosity, and surface complexity could fill these gaps and reduce reliance on physical core samples. Generating such synthetic rocks, however, remains an open and challenging problem, with approaches ranging from generative adversarial networks (GANs) (Mosser et al., 2017, 2018; Zheng & Zhang, 2023) and multi-point statistics (Wu et al., 2018) to process-based models that simulate gravity-driven deposition (Wetzel et al., 2021). GAN-based methods show some promise but are image-trained black-box models, unproven beyond the vicinity of their training data, and also face instability in the training stage along with substantial computational costs in training and generation. We consider thresholding Gaussian random fields (GRFs) (Hyman et al., 2014, 2015) to be the closest existing approach that satisfies the necessary conditions of computational efficiency, interpretability, and statistical control for generating stochastic realizations that mimic porous network structures. Hyman et al. (2015) demonstrated that porous structures generated by thresholding GRFs could reproduce the behavior of porosity and surface area across scales. Hyman and Winter (2014) showed that thresholding GRFs could generate synthetic samples (126³-voxel volume) in good agreement with Berea sandstone sample for the first three Minkowski functionals, and they further proposed a superposition of GRFs to incorporate two correlation length scales for each physical direction.

We build upon the thresholding GRF methodology (Hyman & Winter, 2014) by employing a Matérn covariance kernel, which provides additional control over surface roughness and tortuosity, and integrate this into a multi-threshold framework that segments pores, clay-like regions, and quartz grains into distinct component classes. By treating the clay-like material as a separate class, we gain further insights into its role in the sandstone structure, aiding in generating synthetic sandstone volumes that match the characteristics of the real sandstone. A directional pruning procedure further refines the pore network, reducing the interpretational challenges associated with highly fragmented or disconnected pore systems. We also employ circulant embedding to efficiently sample realizations of the GRF over large domains representative of high-resolution micro-CT volumes, demonstrated here on a 750³ -voxel domain. While thresholded GRFs with low regularity tend to enhance pore network expressiveness, they also amplify interface roughness and introduce artifacts. To mitigate this, one can morphological filtering via a closing operation, which softens surface roughness, bridges narrow gaps, and removes localized noise while preserving the integrity of the larger-scale pore network.

We validate our stochastic generation of synthetic digital rocks against a high-resolution micro-CT scan of a Berea sandstone volume, available from the Digital Rocks Portal (Neumann et al., 2021). We find that the proposed double-thresholded GRFs with a pruned pore network reasonably reconstruct important structural characteristics of the Berea sandstone sample, as demonstrated by the realizations showing overall agreement in the Minkowski functionals, slice-wise pore size distribution, and path-wise tortuosity distribution along the chosen principal direction.

This paper is organized as follows: We begin by describing the micro-CT dataset of Berea sandstone, along with the segmentation procedures and data analysis. We then introduce the pore network modeling and pruning steps, followed by the stochastic modeling approach, detailing the GRF-based generation, double-thresholding, and morphological filtering. A methodology for sampling synthetic micro-CT volumes is presented. The numerical validation section then outlines the quantitative assessment metrics and the experimental setup. We then present validation results comparing synthetic and real Berea sandstone samples for a broad range of digital rock metrics. Finally, we summarize the main insights.

High-Resolution micro-CT Imaging and Sandstone Microstructural Features

This study uses high-resolution 3D micro-computed tomography (micro-CT) image of a Berea sandstone sample, available from the Digital Rocks Portal (Neumann et al., 2021). The volumetric dataset of grayscale values consists of a 1000^3 isometric voxel grid with a voxel spacing of $2.25\ \mu\text{m}$. The scans were acquired using a Bruker SkyScan 1272 micro-CT system; see (Lucas-Oliveira et al., 2022) for additional details. We present a 3D grayscale view of its full 1000^3 -voxel volume in Fig. 1.

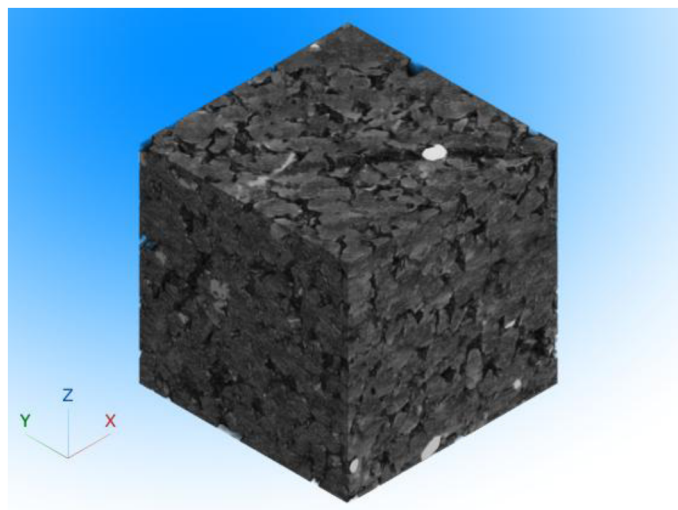


Figure 1—3D grayscale view of the 1000^3 -voxel volume for the Berea sandstone sample: structure is relatively uniform with quartz grains and moderate pore contrast; some large bright inclusions indicate heavy minerals.

The grayscale volume provides insight into the internal heterogeneity. Each voxel intensity reflects the local X-ray attenuation coefficient, which is governed by the material's density and atomic composition. Brighter regions indicate highly attenuating, typically denser materials; black areas correspond to pore spaces, while dark-gray regions generally represent lower-density solid components. Medium-gray intensities primarily represent the quartz grains forming the connected solid matrix. To visualize these features, we present 2D cross-sectional planes, referred to as slices, through the 3D volume taken at the midpoint along each principal direction (X, Y, Z); as shown in Fig. 2. The grayscale variation in the volume, which reflects composition and structure, is analyzed in subsequent sections to support material identification and classification.

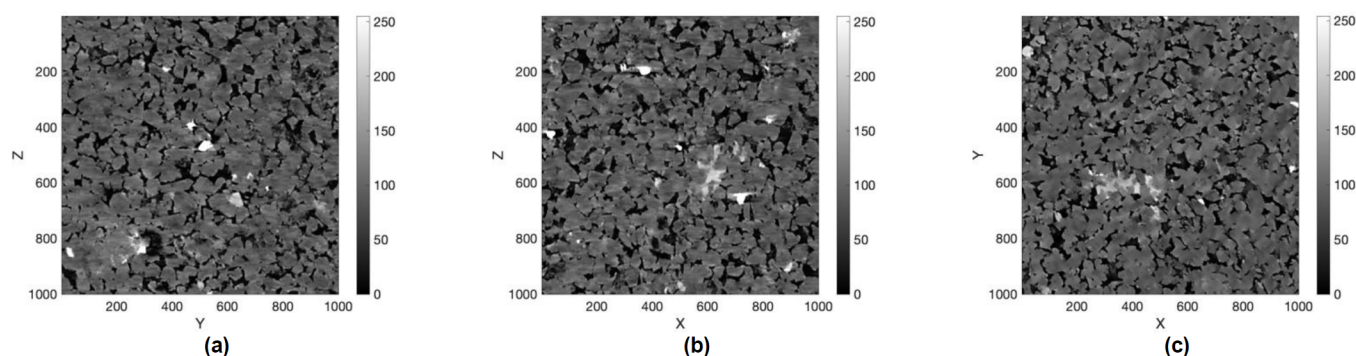


Figure 2—Midpoint 2D slice from the 1000^3 -voxel Berea sandstone micro-CT volume, visualized along each principal direction. Each voxel grayscale encodes X-ray attenuation, with darker regions indicating pore spaces and brighter regions corresponding to typically denser inclusions such as heavy minerals. The bulk of the volume exhibits medium-gray intensities, representing the quartz matrix. (a) X-direction; (b) Y-direction; (c) Z-direction.

The cross-sectional slices show a rather uniform arrangement of similarly sized quartz grains with moderately tight packing, resulting in medium-sized pores. At the highest grayscale levels, small bright patches correspond to heavy minerals or dense quartz inclusions (e.g., pyrite, iron-rich regions), which are sparse and disconnected compared to the continuous quartz matrix. This qualitative interpretation supports using sandstone models that distinguish not only pores versus solid matrix, but also the clay-like material of intermediate grayscale values. The grayscale heterogeneity observed thus motivates an analysis of attenuation-informed material differentiation.

Material Classification and Segmentation Based on Grayscale Analysis

For multi-class segmentation, we conduct localized porosity gradient assessments, placing separation thresholds at statistically significant transitions within the grayscale distribution of the micro-CT volume. This enables a tailored classification that distinguishes not only pores and quartz grains but also transitional regions such as clay-like material, which would otherwise be absorbed into a binary pore–solid representation. By capturing these transitional grayscale ranges, the approach provides further information to subsequent stochastic modeling and structural analysis.

Porosity Gradient-Based Statistical Analysis of Grayscale Transition Zones

To identify physically meaningful grayscale thresholds, we sampled hundreds of 64^3 -voxel subvolumes from each 1000^3 -voxel micro-CT volume and computed porosity across the grayscale range $[4, 255]$. For each threshold increment, T to $T+1$, where T denotes the grayscale cutoff used for binarization, we evaluated the corresponding porosity change $\Delta\phi$ by looking at shifts in the empirical cumulative distribution function (CDF). We construct kernel density estimates (KDEs) of these $\Delta\phi$ values at each threshold, so that we can observe how consistently subvolume porosity responds to incremental increase in T .

By incrementally increasing the tolerance T , one can identify local subvolume porosity deviations from the background trend, which reveal compositional transitions, as illustrated in Fig. 3. The red curves in each plot represent the strict lower and upper bounds observed $\Delta\phi$ across the subvolumes, further highlighting the spatial heterogeneity of the dataset. Based on the local empirical distribution of porosity with respect to T , observed in (a), we identify the following threshold triplet, $T_1 = 25$, $T_2 = 52$, and $T_3 = 230$. These thresholds segment the sandstone volume by grayscale into a classification map of distinct materials: grayscale values below T_1 capture the open pore volume; values in $[T_1, T_2)$ represent clay-like regions; values in $[T_2, T_3)$ represent the quartz matrix; and values $\geq T_3$ are associated with heavy minerals. Fig. 3 (b) presents the grayscale histogram for the Berea sandstone with the normalized frequency of each grayscale value, with colored overlays depicting the segmentation defined by T_1 , T_2 and T_3 . The CDF within each class-defined range provides an estimate of the volume fraction of that class component.

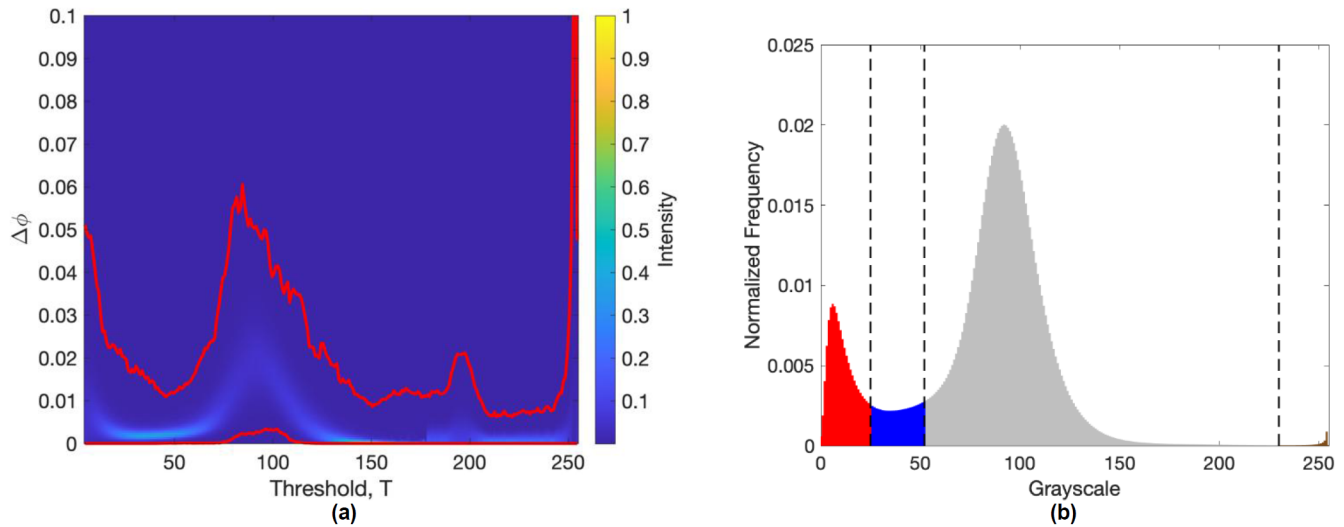


Figure 3—Segmentation and classification by a gradient-based porosity analysis using grayscale data of the Berea sandstone volume. (a) KDEs of incremental porosity change $\Delta\phi$, vertically aligned, across grayscale thresholds, capturing variability in empirical CDF behavior over 64^3 -voxel subvolumes of the original sandstone volume (size 1000^3). Red curves mark the observed minimum and maximum change per threshold increment. Strong local deviations from the background trend identify grayscale intervals associated with different sandstone components; **(b)** The normalized grayscale histogram of the sandstone volume, colored by segmented grayscale classes for the different components. The three vertical lines represent the threshold positions $T_1 = 25$, $T_2 = 52$ and $T_3 = 230$, respectively. Red marks open pore volume, blue marks clay-like material, gray marks quartz matrix, and brown highlights heavy mineral inclusions.

Cumulative distribution within each class-defined range constitutes the volume fraction of each component class. A summary of these statistics for both samples provided in [Table A](#).

Table A—Estimated volume fraction for each class based on our grayscale-based segmentation using thresholds (T_1 , T_2 , T_3). Values represent the proportion of voxels in each threshold-defined grayscale interval for the Berea with triplet ($T_1 = 25$, $T_2 = 52$, $T_3 = 230$).

Segmentation Class	Grayscale Interval	Berea
Pores	$[0, T_1)$	0.1313
Clay-like minerals	$[T_1, T_2)$	0.0634
Quartz	$[T_2, T_3)$	0.8028
Heavy minerals	$[T_3, 255]$	0.0024

Thresholding forms the basis for our stochastic modeling framework presented later for generating digital sandstone samples, ensuring statistical consistency between the real micro-CT volume and the stochastically generated realizations. The distribution mass above T_3 , corresponding to heavy mineral inclusions, is relatively small. Thus, we restrict our attention to modeling the structural and statistical features of the open pore volume, the clay-like material, and the quartz matrix, whereas incorporating heavy mineral inclusions is left as a natural future extension of the model. The heavy minerals account for less than 0.3 % of the total volume, confirming its volumetric insignificance, see [Table A](#). Future work is also needed to distinguish between quartz and other materials that may reside within the grayscale range $[T_2, T_3)$, such as denser clays and feldspar, which are currently classified as quartz in our segmentation. These assumptions could be refined within our framework; however, modeling such inclusions of such materials would likely require additional stochastic sub-model components tailored to each material type. The threshold T_3 is still essential for accurately defining the upper bound for the quartz matrix.

[Figure 4](#) illustrates how thresholding performs in identifying different sandstone components, where we overlay the 2D midpoint slices (shown in [Fig. 2](#)) with red contours tracing connected pore regions with

grayscale values below T_1 , and blue dots indicating voxels in the intermediate range $[T_1, T_2)$, interpreted as clay-like material. More highly attenuating materials than threshold T_2 are not marked.

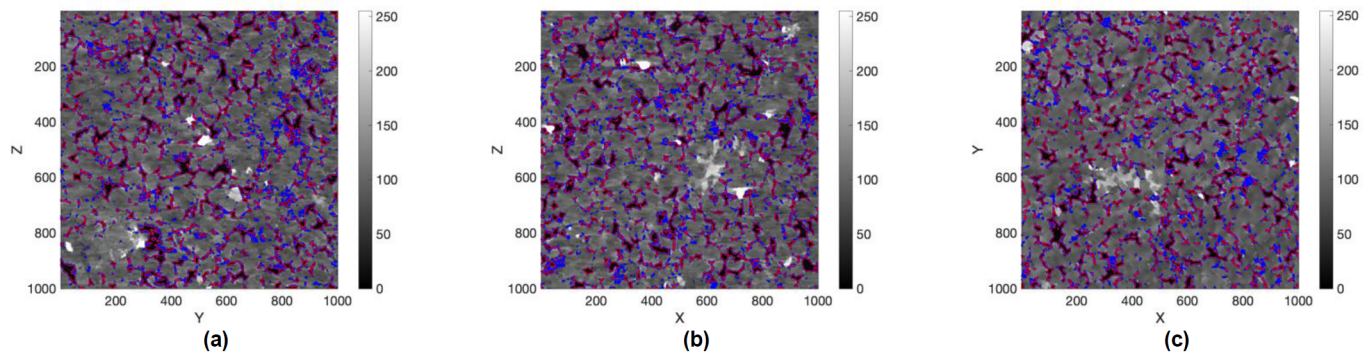


Figure 4—Segmentation of midpoint 2D slices of the sandstone volume using thresholds $T_1 = 25$ and $T_2 = 52$. Red contours highlight connected pore regions with grayscale values below T_1 , while blue dots mark voxels in the clay-like material range $[T_1, T_2)$. (a) X-direction; (b) Y-direction; (c) Z-direction.

The clay-like material frequently appears along pore boundaries or as thin laminations (Kareem et al., 2017), as shown in Fig. 5 on the 150^3 -voxel scale. This multi-scale pattern motivates our use of a multi-class segmentation approach rather than a simple binary classification, and supports interpreting the clay-like material as a transitional component between pores and the quartz matrix.

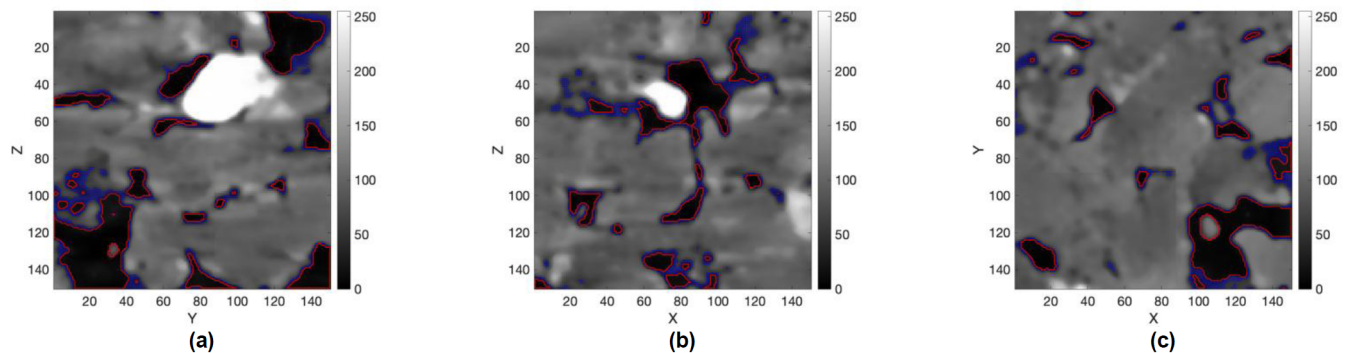


Figure 5—Segmentation of midpoint slices extracted from centered 150^3 -voxel subvolume of the Berea. Red contours highlight connected pore regions, while blue dots indicate voxels identified as clay-like material, supporting the observation that, on the finer scales, clay-like material is a transitional component between pores and quartz grains. (a–c) Slices in the X-, Y-, and Z-directions, respectively.

Pore Network Extraction and Modeling

For the pore network extraction, we begin by structuring the 3D sandstone sample as a stack of 2D cross-sectional slices along a selected principal direction (X, Y, or Z). Within each slice, we identify clusters, defined as closed sets of voxels satisfying a grayscale threshold T , using MATLAB's built-in `bwconncomp` function with 8-connectivity. Here, 8-connectivity reflects that voxels are connected under the condition of sharing a face, edge, or a corner, which is appropriate for microstructures where tortuous paths are common. Two clusters in adjacent slices are considered locally connected if they share at least one voxel position, forming a local cluster-to-cluster overlap across slices.

More specifically, each 2D slice is processed independently to extract all clusters, where each cluster is stored in a structured array that includes its centroid coordinates, interior and boundary voxel locations, and pointers to cluster-to-cluster connections in both forward and backward directions, enabling consistent bidirectional traversal. If local cluster-to-cluster overlaps persist from one side of the volume to the other

along the chosen direction, they form a *spanning path*. The active status of each cluster is tracked during a forward traversal from the initial boundary slice, while any deactivated cluster (a dead end) propagates this information backward to recursively deactivate clusters not contributing to any spanning path. Based on the characterized pore network, we apply a directional filtering step in which clusters not contributing to any spanning path are removed; this is hereafter referred to as *directional pruning*. The idea is similar to 3D percolation-based filtering in pore network modeling, where the largest spanning cluster is retained to represent the primary flow-conducting structure, as in (Dong & Blunt, 2009). The pruning preserve pathways spanning from inlet to outlet, preserving a percolating structure relevant for flow. Figure 6 presents the raw centered 500^3 -voxel subvolume for Berea, together with visualizations of the corresponding pruned skeleton and classified volumes, highlighting pore network connectivity and material composition, which are further analyzed in the next section.

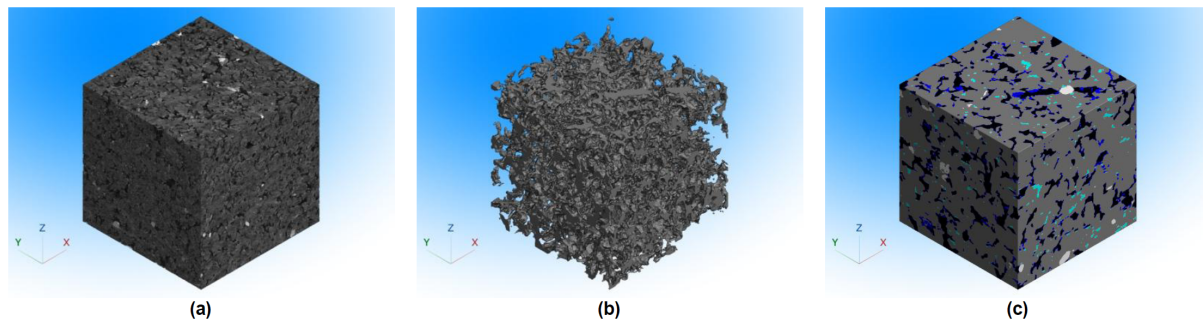


Figure 6—3D visualization of a centered 500^3 -voxel subvolume of the Berea sandstone volume. The left image shows the raw grayscale volume, the middle image displays the pruned pore network skeleton with cyan highlighting regions trimmed along the Z-direction using threshold T_2 , and the right image presents the classified volume segmented into pores (black), clay-like material (blue), quartz grains (gray), and heavy mineral inclusions (white). (a) Raw volume; (b) Pruned skeleton; (c) Classified volume.

By rotating the original volume and repeating the same extraction strategy along other Cartesian axes, one can systematically study directional anisotropy in pore structure. This framework enables the assessment of geometric tortuosity, directional pore size distributions, and effective porosity in heterogeneous samples, and could also be integrated with digital rock-based transport modeling.

Data Analysis of Sandstone Volumes

This section presents numerical results comparing pore network characteristics with and without directional pruning. The aim is to investigate how pruning affects network connectivity and structure, and to support the assessment of flow-relevant features. More specifically, the goals are to assess the influence of directional pruning on slice-wise pore size distributions; analyze pore size distributions along the X, Y, and Z directions; assess directional geometric tortuosity in conjunction with pore sizes, which together may reveal patterns in anisotropic transport behavior; and examine the effect of including or excluding clay-like material in the initial pore classification, corresponding to using threshold T_1 or T_2 , respectively, in the grayscale-based thresholding segmentation. *Slice-wise pore size distribution*, denoted by $P(r)$, represents the distribution of pore radii in planes orthogonal to a specified principal direction. This is based on the simplifying assumption of circular pore geometry. In practice, it is obtained by measuring the area (voxel count) of each individual pore cross section, translating it into an equivalent radius, and then aggregating these radii across all slices. This captures spatial variability and may reveal any structural directional anisotropy.

Furthermore, we analyze network complexity through the *centroid-to-centroid path-wise tortuosity distribution*, denoted by $P(\tau)$. This distribution is estimated via equally-likely path sampling, given the vast number of viable routes. Paths start from randomly selected pore clusters on the inlet boundary and proceed through cluster-to-cluster connections along the specified direction. At each branching point, when multiple

overlapping clusters appear in the next slice, we draw uniformly to select the next step, rather than favoring minimal resistance. The accumulated normalized centroid-to-centroid distance from inlet to outlet facet defines the tortuosity of each sampled path. Unlike methods that compute only the minimum normalized path using algorithms such as Dijkstra or A* (Eloisa Borello et al., 2022), our approach quantifies tortuosity variability across the spanning paths, capturing both the dominant behavior and its spread; the conventional minimum merely reflects the extreme lower tail. By this approach, we characterize the directional pore network of the Berea sandstone, with numerical results shown in Fig. 7.

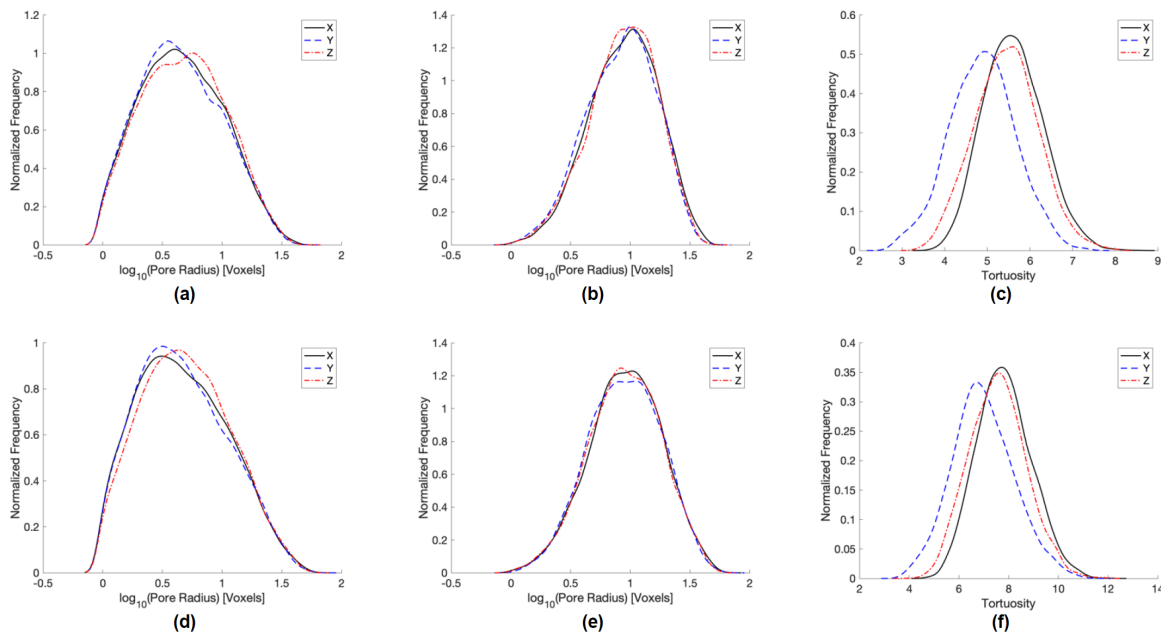


Figure 7—KDE distributions for the Berea sandstone showing slice-wise pore size distribution $P(r)$ and centroid-to-centroid path-wise tortuosity distribution $P(\tau)$ along all three principal axes. When using threshold T_1 (open pores), we have directional distributions: (a) $P(r)$; (b) $P(r)$ in pruned network; (c) $P(\tau)$ in pruned network. With T_2 (joint volume of pores and clay), (d) $P(r)$; (e) $P(r)$ in pruned network; (f) $P(\tau)$ in pruned network. We empirically estimate tortuosity distribution $P(\tau)$ from 5,000 random samples, as motivated earlier in this section.

From Fig. 7, comparing the first column (original network) with the second (pruned network) shows that after pruning, the directional slice-wise pore size distributions $P(r)$ become distinctly log-normal and appear approximately isotropic. Furthermore, (c,f) reveals that the path-wise tortuosity $P(\tau)$ is lowest along the Y-direction, despite visually similar pore size distributions across all directions, as seen in (b,e). The pores relevant for the Y-direction lie on the XZ-cross-sections of the pruned volume and exhibit a modest leftward shift in their radii distribution (e,f), which restricts transverse flow and thus leads to lower tortuosity along Y. Even subtle anisotropy in these distributions proves meaningful for establishing the observed ordering in tortuosity.

Stochastic Modeling of Pore Network in Sandstones

Early studies by Adler and Taylor (2009) on excursion set geometry in random fields, and by Torquato (2002) on effective properties and spatial statistics of idealized heterogeneous media, introduced key concepts relevant to thresholding GRFs. Although not originally developed for digital rocks or porous media, this approach was later adapted for digital rock modeling by Hyman et al. (2014, 2015), who employed thresholding GRF to generate synthetic pore structures and examined the sensitivity of morphological statistics to correlation length. Their work demonstrated the potential to reproduce a wide range of morphologies using a single threshold, focusing primarily on scaling behavior and Minkowski functionals. Combined with a double-thresholding scheme, our approach extends the generative framework

to include transitional clay-like material and allow higher flexibility in matching sandstone sample data. Hyman et al. (2015) studied how geometric and topological features of thresholding GRF-based structures scale with volume size, reporting evidence of multifractal behavior across scales.

The GRF serves as a latent stochastic process from which microstructural classes (e.g., pore, clay-like, and quartz) can be inferred via thresholding. The standard Matérn GRF, detailed in (Rasmussen & Williams, 2005), is described in the next section, while the thresholding procedure and its connection to generating digital rocks consistent with the underlying grayscale-based segmentation are discussed thereafter.

Matérn Gaussian Random Field

Let $Z(x)$ be a real-valued Gaussian random field (GRF) with a stationary covariance function defined over a spatial domain $D \subset \mathbb{R}^3$. We generate realizations of $Z(x)$ on a regular, structured grid of size $L \times L \times L$ with the spatial locations given by $\{x_i\}_{i=1}^N \subset D$ where $N = L^3$. The random field $Z(x)$ at those locations is given by

$$Z = (Z(x_1), \dots, Z(x_N))^T \in \mathbb{R}^N,$$

which follows a multivariate Normal with zero mean and covariance matrix denoted by $\Sigma \in \mathbb{R}^{N \times N}$, with entries $\Sigma_{ij} = C(x_i - x_j)$ for a given stationary covariance kernel C . The GRF is specified by its mean and covariance function C , and we use the Matérn covariance kernel, which defined as

$$C(h) = \sigma^2 \frac{2^{1-\nu}}{\Gamma(\nu)} (\sqrt{2\nu} \|h\|)^{\nu} K_{\nu}(\sqrt{2\nu} \|h\|),$$

where $h = x - x'$ is the distance between two locations x and x' in D ; Γ is the Gamma function; K_{ν} is the modified Bessel function of the second kind; $\sigma^2 (> 0)$ is the marginal (process) variance controlling the random fluctuations; $l_x, l_y, l_z (> 0)$ are the correlation lengths along each spatial axis, for which higher values translate to longer correlation reach in the spatial domain D ; $\nu (> 0)$ controls the regularity (smoothness) of the realizations of the GRF; and $\|h\|$ denotes a generalized anisotropic norm:

$$\|h\| = \sqrt{\left(\frac{h_x}{l_x}\right)^2 + \left(\frac{h_y}{l_y}\right)^2 + \left(\frac{h_z}{l_z}\right)^2}.$$

If l_x, l_y, l_z are equal, we attain an isotropic field, exhibiting statistically identical behavior in all directions, whereas in our formulation we allow general anisotropy by allowing them to be mutually independent. The smoothness parameter ν governs the smoothness level of the resulting realizations, with larger values producing smoother field realizations. For example, $\nu = 0.5$ corresponds to an exponential kernel that generates rough, non-differentiable fields, while $\nu = \infty$ yields the infinitely differentiable squared exponential kernel. Intermediate values such as $\nu = 1.5$ or $\nu = 2.5$ produce fields that are once or twice mean-square differentiable, respectively, and are commonly used in geostatistical applications. The appropriate choice of ν depends on the level of spatial continuity desired, and can be treated as a tuning parameter. The positive-definite nature of the Matérn covariance ensures that valid multivariate normal realizations can be generated over the grid.

In this work, we draw realizations of GRFs on our regular, structured 3D domain using the circulant embedding method (CEM) (Dietrich and Newsam, 1997), which constructs a periodic extension of the covariance function. CEM generates exact-in-distribution samples for structured grids and stationary covariance models (Liu et al., 2019), offering a computational work complexity of $O(N \log(N))$ by circumventing the need to form the full covariance matrix Σ . However, to ensure the validity of the embedding, the domain must be sufficiently large for the embedded covariance matrix to remain positive definite. In our setup, we generate a larger domain and perform the analysis on an interior subvolume, maintaining a conservative buffer zone. Hyman and Winter (2014) previously used convolution-based sampling for thresholded GRFs with a squared-exponential covariance, which is infinitely differentiable.

This approach has the same work complexity as CEM but is only approximate due to kernel truncation and can be less straightforward for controlling directional scaling and regularity.

Thresholding GRFs for Pore Network Segmentation

Thresholding GRFs in porous media has typically focused on binary segmentation, where Gaussian random fields (GRFs) are thresholded to stochastically generate synthetic porous networks (Hyman et al., 2014, 2015). In this framework, a GRF $Z(x)$ defined over a spatial domain is used to assign pore-solid classifications based on excursions above or below a given threshold, usually set to achieve a target volume ratio. A single threshold T classifies regions where $Z(x) < T$ as pore space and the remainder as solid. This can be viewed as constructing a level set of the GRF, which is straightforward under Gaussianity and stationarity assumptions (Adler & Taylor, 2009).

Double Thresholding using Quantiles

Real geological materials often contain transitional phases, such as clay films, that cannot be adequately captured using a single threshold. To address this, we introduce a *double-thresholding* model in which the GRF $Z(x)$ is partitioned into three material classes using two thresholds:

$$T_1 = F^{-1}(q_1), \quad T_2 = F^{-1}(q_2),$$

where F^{-1} is the inverse cumulative distribution function (quantile function) of the standard normal distribution, and $0 < q_1 < q_2 < 1$. The resulting thresholding GRF for the classification, denoted by $Z_{\text{class}}(x)$, is defined as: $Z_{\text{class}}(x) = 1$, if $Z(x) < q_1$ (pores); $Z_{\text{class}}(x) = 2$, if $q_1 < Z(x) \leq q_2$ (clay-like minerals); and $Z_{\text{class}}(x) = 3$, if $q_2 < Z(x) \leq 1$ (quartz). Thresholds are defined via target volume fractions, using quantiles q_1 and q_2 of the GRF's marginal distribution. Since classification depends only on the relative ranking of GRF values, the mean and variance of the GRF do not affect the resulting segmentation. In Berea sandstone, clay minerals such as kaolinite and illite make up only a small fraction of the bulk volume, yet often appear as thin, spatially coherent films along pore boundaries, diminishing pore connectivity (Kareem et al., 2017). Although the present model does not distinguish between different shapes and spatial arrangements of the clay-like material (e.g., coatings versus occlusions), it provides a step toward reproducing the composite nature of real sandstone. Future work may build on this by introducing additional latent random fields or conditional dependencies to better capture such heterogeneity. This is the motivation for employing double-thresholding here: the intermediate value range between two thresholds is defined as a transition zone, which in our case models the transition from pores to quartz grains through clay-like material.

Morphological Filtering of Thresholded GRF Classification

Thresholded GRFs often exhibit a trade-off: low Matérn smoothness ν produces overly rough boundaries, while high ν leads to unrealistically smooth shapes. Since sandstone networks can feature highly tortuous paths that require low ν to capture, we apply morphological filtering to gain an extra degree of freedom in controlling boundary shape and regularity, and to suppress spurious artifacts when needed. Morphological closing refines the boundaries of complex pathways and fills minor gaps, while preserving the directional features and connectivity governed by the GRF. This enhances the realism of the network in line with two Minkowski functionals: specific surface area and specific mean breadth. Starting from the material classification obtained via double-thresholding the GRF (where labels 1 and 2 correspond to pore and clay-like regions), we construct a binary mask by assigning 1 to voxels with labels 1 or 2 and 0 otherwise. MATLAB's built-in `imclose` function is then applied with a cubic structuring element of size L_{close} . All voxels outside the filtered mask are reassigned to class 3 (quartz). This process does not alter the underlying GRF model; it serves as a localized, shape-level refinement of the final classification. Compared to convolutional smoothing, morphological filtering preserves sharp interfaces, yielding more polygonal, rock-like features.

Stochastic Generation of Synthetic Micro-CT Sandstone Data

To synthesize realistic grayscale variation within each classified component, we construct class-specific inverse mappings from normalized Gaussian random fields to grayscale values by inverting the empirical cumulative histograms of the real micro-CT data restricted to each class. The global grayscale distribution is truncated at a high-intensity threshold T_3 to exclude heavy mineral inclusions, which are not targeted in the stochastic modeling presented here. Using the threshold triplet (T_1, T_2, T_3) , we segment the grayscale range into three disjoint intervals corresponding to the component classes, ensuring that each phase retains its empirically observed grayscale intensity. Each segment of the empirical CDF is then renormalized and inverted using piecewise-linear interpolation, producing mappings

$$F_j^{-1}: [0, 1] \rightarrow \mathbb{R} \text{ for } j \in \{1, 2, 3\}$$

where 1, 2, and 3 correspond to pore, clay-like minerals, and quartz, respectively. Synthetic class-based grayscale fields are generated by applying a transformation to independent GRFs $Z_j(x)$ defined by:

$$G_j(x) = F_j^{-1}(\Phi_N(Z_j(x)))$$

where Φ_N is the standard normal CDF. The final grayscale field, denoted by $Z_{\text{grayscale}}(x)$, is defined on the domain D with values given by patching together synthetic class-based grayscale fields $G_j(x)$, $j \in 1, 2, 3$, on non-overlapping regions in D : $Z_{\text{grayscale}}(x) = G_1(x)$, if $Z_{\text{class}}(x) = 1$ (pore region); $G_2(x)$, if $Z_{\text{class}}(x) = 2$ (clay-like region); $G_3(x)$, if $Z_{\text{class}}(x) = 3$ (quartz region), depending on the GRF $Z_{\text{class}}(x)$. To match the spatial statistics of the real CT data, we estimate the correlation lengths $l_{x,j}$, $l_{y,j}$, $l_{z,j}$ and marginal variance σ_j^2 for each class $j = \{1, 2, 3\}$ based on the grayscale volume. Grayscale intensities are first mapped to standard normal scores by inverting the global histogram and applying the normal quantile function. We adopt Matérn GRF for G_j over the domain D and with regularity parameter $\nu = 2.5$.

Numerical Validation Against Micro-CT Data for a Berea Sandstone

This section evaluates the fidelity of the proposed stochastic modeling framework by comparing synthetic realizations against the Berea sandstone sample, using a publicly available high-resolution 1000³-voxel micro-CT volume dataset (spacing 2.25 μm) from the Digital Rocks Portal (Neumann et al., 2021). For the comparison, we use an extracted centered volume of 500³ voxels from the dataset. The validation focuses on structural, morphological, and topological metrics that reflect pore-scale behavior relevant to flow and transport modeling.

Validation Procedure: Minkowski Functionals and Distribution-based Metrics

To assess the consistency of the generated synthetic rock volumes with real micro-CT sandstone data, we employ geometric and topological metrics commonly used in digital rock physics, namely the classical Minkowski functionals, to characterize pore-scale structure. To complement these, we introduce two distribution-based measures of the pore network: the centroid-to-centroid path-wise tortuosity distribution, $P(\tau)$, and the directional slice-wise pore size distribution, $P(r)$, as detailed in the data analysis section. The Z-direction acts as the principal direction in pruning the network. The Minkowski functionals provide volumetric and topological invariants that describe the shape and connectivity of the pore space independently of explicit network models. We compute these using a Crofton-based estimation method implemented in the Python package `quantimpy.minkowski` (Boelens et al., 2021), which evaluates curvature-based integrals on voxel-based binary volumes. The four normalized functionals per unit volume are: porosity ϕ (unitless), surface area S_V (μm^{-1}), mean breadth B_V (μm^{-2}), and the Euler characteristic χ_V (unitless). These functionals together provide a comprehensive characterization of the network in terms of its volume fraction, surface complexity, average convexity, and topological connectivity, respectively. For

further details on these classical Minkowski functionals and their estimators, see (Armstrong et al., 2019; Boelens et al., 2021). Note that effective porosity is naturally obtained from the pruned network, since only pore clusters forming spanning pathways remain.

Experimental Setup

A synthetic volume is generated on a 750^3 -voxel domain using our stochastic double-thresholding GRF, with multiple realizations drawn under identical GRF settings. and this is repeated under the same setting of the GRF to attain multiple realizations. For each synthetic volume realization, we extract a centered subvolume of 500^3 voxels. Summary statistics of the validation metrics across the realizations are presented, along with visualization of individual outcomes for the pore size and tortuosity distributions. For the double-thresholding Matérn GRF, we specify quantiles $q_1 = 0.135$ and $q_2 = 0.2$, corresponding to porosities associated with the thresholds ($T_1=25$, $T_2=52$) for pore-clay classification. The correlation parameters are set to $l_x = 12.5$, $l_y = 14.5$, $l_z = 14.5$ with a regularity of $\nu = 1.7$. To further refine the structure, we apply the morphological closing discussed previously, using a closing operation with a cubic element of width 3 voxels. We prune the network for T_2 along the Z-direction.

All computations are performed in MATLAB R2024a with the Parallel Computing Toolbox to accelerate large 3D volume processing and batch evaluations. Minkowski functionals are computed using the QuantImpy package (version 0.4.6) for Crofton-based curvature and topology computations, called via MATLAB's Python environment (pyenv).

Numerical Demonstration

We draw eight realizations of the double-thresholding GRF using the experimental setup described, and present their aggregated statistical summaries for the mean and standard deviation of the Minkowski functionals in Table C, for grayscale binary segmentation with cutoff threshold T_1 , and in Table D with cutoff T_2 , alongside the corresponding values from the micro-CT of the Berea sandstone. The synthetic digital rocks show agreement in the mean with the Minkowski functionals of the real Berea sandstone for most metrics, indicating that the thresholded GRF approach effectively captures essential structural characteristics. The standard deviations are of a reasonable magnitude, suggesting that the method maintains control over the statistical properties while still reflecting natural variability. The GRF settings were selected based on interpretations of its parameter influence, followed by parameter refinement.

Table C—Comparison in Minkowski functionals for Berea *pruned* pore network from thresholding $T < T_1$ on 500^3 -voxel centered subvolume against synthetically generated pore networks. The relative difference is computed as the percentage deviation of the synthetic mean from the real value. Metrics include effective porosity, surface area to volume ratio, and mean breadth, per unit volume, and Euler characteristic.

Metric Quantity	Real Value	Synthetic Mean	Relative Difference	Standard Deviation
Effective Porosity (ϕ_{eff})	0.116	0.111	4.9 %	0.00395
Surface Area (S_V)	3.14	3.15	0.5 %	0.0950
Mean Breadth (B_V)	$3.85 \cdot 10^{-8}$	$5.17 \cdot 10^{-8}$	34 %	$1.26 \cdot 10^{-9}$
Euler Characteristic (χ_V)	-300	-50	83 %	30.4

The primary outliers occur for T_1 in the mean breadth and Euler characteristic. This suggests that modeling the open pore volume alone is more challenging, especially for connectivity and boundary regularity, likely due to clay-like inclusions accumulating along pore walls. With T_2 (joint pore-clay network), the structure becomes more predictable and meets a broader set of criteria (see Table D).

Table D—Comparison in Minkowski functionals for Berea *pruned* pore network from thresholding $T < T_2$ on 500³-voxel centered subvolume against synthetically generated pore networks. The relative difference is computed as the percentage deviation of the synthetic mean from the real value. Metrics include effective porosity, surface area to volume ratio, and mean breadth, per unit volume, and Euler characteristic.

Metric Quantity	Real Value	Synthetic Mean	Relative Difference	Standard Deviation
Effective Porosity (ϕ_{eff})	0.164	0.160	2.3 %	0.00549
Surface Area (S_V)	3.75	3.69	1.5 %	0.108
Mean Breadth (B_V)	$3.07 \cdot 10^{-6}$	$3.65 \cdot 10^{-6}$	19 %	$8.03 \cdot 10^{-10}$
Euler Characteristic (χ_V)	-849	-837	1.3 %	40.0

With the thresholded GRF approach, the primary control is over the porosity of the sample to be generated, which is conservatively satisfied in a statistical sense by the specified quantiles. However, this differs from the effective porosity, which drops significantly once the network is pruned. As seen in [Tables C and D](#), we achieve a rather close agreement with the target effective porosity, despite specifying original porosities that are higher, and despite the fact that this drop is not easily predictable. The standard deviation of the effective porosity for the eight realizations is relatively low as well, indicating sufficient control to generate synthetic sandstones near the desired porosity level. This holds both for the porosity reflecting the open pore space volume, and for the combined porosity when pores and clay-like material are treated together.

In [Figure 8](#), we present the individual outcomes of the synthetic realizations compared to the real sandstone sample for the slice-wise pore size distribution, $P(r)$ and the centroid-to-centroid path-wise tortuosity distribution, $P(\tau)$. Overall, the realizations match well in both pore radius and tortuosity for thresholds T_1 and T_2 . The agreement at both thresholds is a strong indicator of the effectiveness of our proposed double-thresholding GRF strategy, demonstrating that the synthetic sandstone realizations not only align with the Minkowski functionals, but also with metrics more directly related to transport.

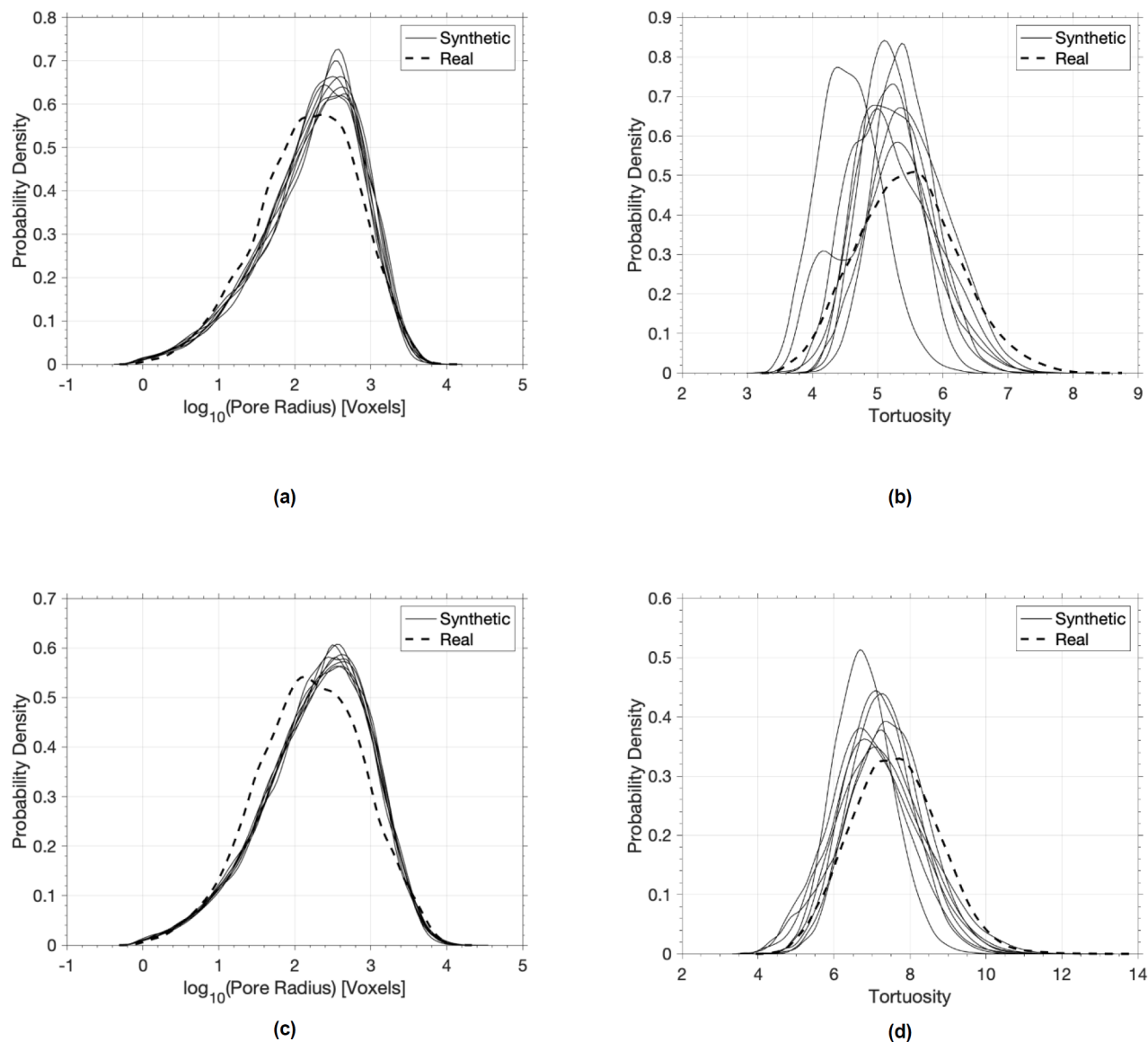


Figure 8—Comparison of pore network characteristics derived from the real Berea sandstone and eight synthetic realizations, emphasizing pore size distributions $P(r)$ and tortuosity $P(\tau)$ under two thresholds, T1 and T2, shown in the first and second rows, respectively. T1 isolates the open pore space, while T2 combines open pores and clay-like material to represent a viable network structure. (a,c) Empirical pore size distribution $P(r)$ of the logarithm of pore radii along the Z-direction (from XY cross sections) using thresholds T1 and T2, respectively; (b,d) Empirical tortuosity distribution $P(\tau)$ along the Z-direction under thresholds T1 and T2, respectively.

Figure 9 illustrates the application of our stochastic generation framework to produce synthetic micro-CT sandstone data for two of the realizations described above. We observe a strong match in the grayscale distributions across the synthetic volumes, as shown in (a), and in (c) and (f) we see the corresponding synthetic realizations replicating the micro-CT Berea sandstone structure. It should be noted that values above grayscale 230 correspond to heavy mineral inclusion.

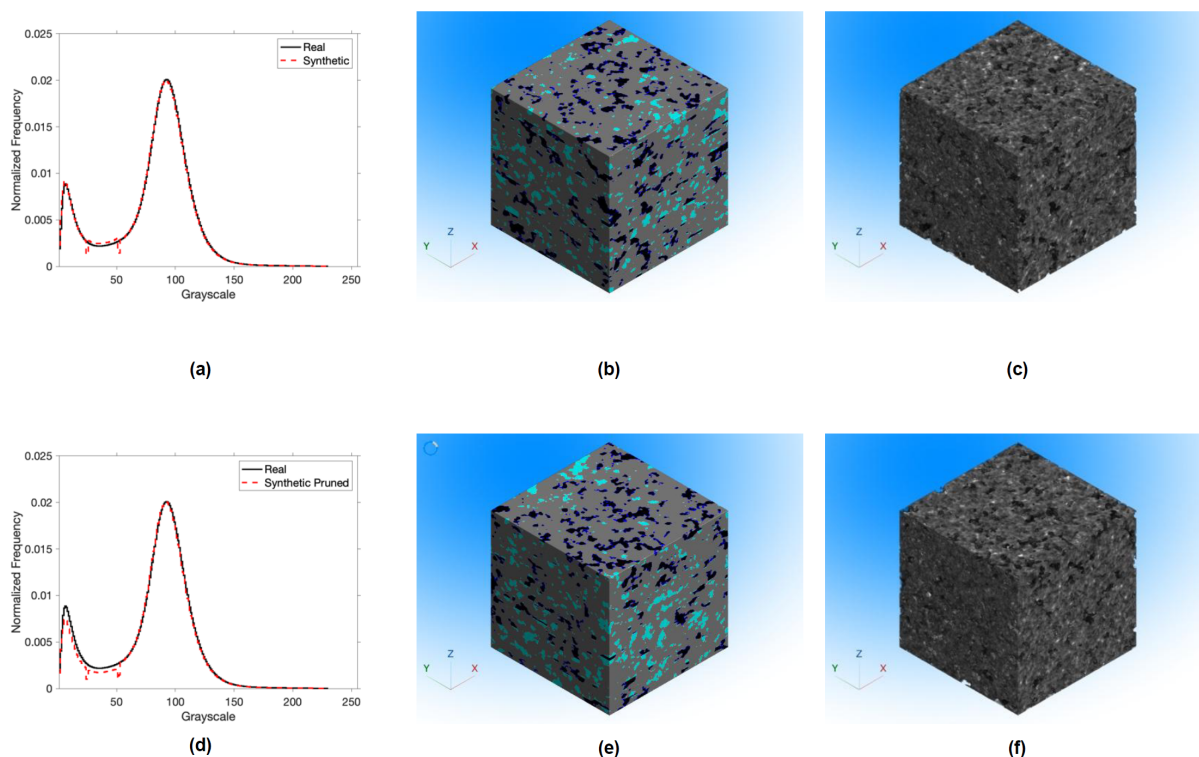


Figure 9—Two synthetic realizations are shown by their grayscale distribution, classification, and the resulting synthetic digital rock structures. The first and second realization are shown in rows 1 and 2, respectively: (a,d) Grayscale histograms of the synthetic realizations versus the real sample; (b,e) Classification into pore (black), clay-like (blue), and quartz (gray), and including removed paths (cyan) as a result of the pruning step; (c,f) Corresponding synthetic digital rocks generated using double-thresholding GRF with grayscale distributions matched to the real data. See Fig. 6 for qualitative comparison with the real Berea sandstone sample.

Conclusions

The numerical validation against the Berea sandstone sample using our double-thresholding GRF strategy show encouraging statistical agreement across multiple synthetic digital rock realizations. The assessment included Minkowski functionals, pore size distributions, and path-wise tortuosity along an envisioned porous flow direction. The standard deviations of these metrics across the synthetic sandstone samples were neither too small nor excessively large, indicating that the thresholded GRF reproduces structures in a controlled manner while still introducing sufficient variability to keep realizations within the expected proximity of the target sandstone characteristics. In this model, the correlation parameters directly influence connectivity, anisotropy, and pore size distributions, while the regularity parameter governs curvature, fragmentation, and surface roughness.

In summary, this work proposed a stochastic modeling framework based on a novel double-thresholding Gaussian random field (GRF) strategy to generate high-resolution synthetic sandstone volumes, with particular emphasis on representing clay-like material at the interfaces between pore volumes and quartz grains. Numerical results for micro-CT of Berea sandstone demonstrate that there exists a setting of the proposed method that produces synthetic high-resolution digital sandstone volumes with reasonable accuracy and robust statistical agreement, as shown by metrics for effective porosity, pore size distributions, surface area, curvature, connectivity, topological features, and tortuosity. In contrast to typical black-box machine learning approaches for synthetic digital rock generation, this framework provides interpretable control over structural attributes through its model parameters, making it a promising option for sensitivity analysis and uncertainty quantification in digital rock analysis and flow simulation studies.

Nomenclature

B_V	Specific mean breadth (Minkowski functional, per unit volume)
$C(h)$	Covariance function of the Gaussian random field (GRF)
D	Sandstone volume domain
G_1, G_2, G_3	Gaussian random fields used to generate the synthetic grayscale texture structure
L	Number of voxels along each spatial direction in the discretized 3D domain of the GRF
L_{close}	Width of cubic structuring element used in morphological closing
N	Number of total voxels in the discretized 3D domain of the generated GRF
$P(r)$	Probability density function of the slice-wise pore radius
$P(\tau)$	Probability density function of the centroid-to-centroid path-wise tortuosity
q_1, q_2	Quantile associated with cutoff thresholds T_1 and T_2 for the grayscale distribution, respectively
S_V	Specific surface area (Minkowski functional, per unit volume)
T_1	Cut-off threshold in grayscale distribution for open pores
T_2	Cut-off threshold in grayscale distribution for open pores and clay-like material
T_3	Cut-off threshold in grayscale distribution for open pores, clay-like material and quartz grains
$Z(x)$	GRF at physical location x in domain
Z_{class}	Classified field after thresholding GRF into pores, clay-like and quartz grain
ϕ	Total porosity (Minkowski functional)
ϕ_{eff}	Effective porosity after pruning of non-spanning paths (disconnected clusters)
χ_V	Specific Euler characteristic (Minkowski functional, per unit volume)
l_x, l_y, l_z	Correlation lengths in the X, Y and Z directions (hyper-parameters in the GRF covariance)
ν	Regularity parameter (hyper-parameter in the GRF covariance)
σ	Standard deviation of GRF

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